Split-Step Fourier Method for the Nonlinear Schr"odinger Equation

The NLS Equation

We consider the cubic nonlinear Schr"odinger equation (NLS):

$$iu_t + \frac{1}{2}u_{xx} + \kappa|u|^2 u = 0$$

on a periodic domain.

Operator Splitting Idea

Write the equation as:

$$u_t = (L + N)u$$

where

$$Lu = i\frac{1}{2}u_{xx} \quad Nu \qquad\qquad = i\kappa|u|^2 u$$

We solve separately:

- Linear part (dispersion)
- Nonlinear part (local phase rotation)

Linear Evolution

In Fourier space:

$$\hat{u}_t = -i\frac{1}{2}k^2\hat{u}$$

Solution:

$$\hat{u}(t + \Delta t) = \hat{u}(t)e^{-ik^2\Delta t/2}$$

Nonlinear Evolution

Pointwise ODE:

$$u_t = i\kappa|u|^2 u$$

Solution:

$$u(t + \Delta t) = u(t)e^{i\kappa|u|^2\Delta t}$$

Amplitude is constant; phase rotates.

Strang Splitting

Second-order accurate splitting:

$$e^{\Delta t(L+N)} \approx e^{\frac{\Delta t}{2}N} e^{\Delta t L} e^{\frac{\Delta t}{2}N}$$

Procedure:

1. Half nonlinear step
2. Full linear step
3. Half nonlinear step

Algorithm (One Time Step)

Given u^n:

1. $u \leftarrow u^n e^{i\kappa|u^n|^2\Delta t/2}$
2. FFT: $u \rightarrow \hat{u}$
3. $\hat{u} \leftarrow \hat{u}e^{-ik^2\Delta t/2}$
4. IFFT: $\hat{u} \rightarrow u$
5. $u^{n+1} = ue^{i\kappa|u|^2\Delta t/2}$

Properties

- Spectral accuracy in space
- Second-order accuracy in time
- Efficient: $O(N \log N)$ per step
- Preserves qualitative features (mass, approximate energy)

Physical Interpretation

- Linear step: dispersion spreads wave packet
- Nonlinear step: phase modulation (self-focusing/defocusing)
- Interaction produces solitons, modulation instability

Summary

- Split-step Fourier method separates physics
- Exact solution of each subproblem
- Strang splitting gives second-order accuracy

Time Step Considerations

Each substep is solved exactly, splitting introduces error.

Guidelines:

- Time step Δt must resolve nonlinear phase rotation
- Require $\kappa |u|^2 \Delta t \ll 1$
- Also resolve highest Fourier mode: $k_{max}^2 \Delta t \ll 1$

In practice:

- Accuracy (not stability) is the main restriction
- Choose Δt small enough to resolve both effects

Aliasing and Resolution

- Nonlinearity generates higher frequencies
- Need sufficient spatial resolution
- Use dealiasing if necessary (2/3 rule)

Poor resolution can mimic instability.

Soliton Initial Condition

For focusing NLS ($\kappa > 0$), a soliton solution:

$$u(x, 0) = A, \operatorname{sech}(Ax)$$

Exact solution propagates without changing shape.

What to Observe

- Shape preservation of soliton
- Small phase error over long times
- Sensitivity to Δt and resolution